STRATEGIC ALIGNMENT BRIEF · TRANSITION FINANCE & NATURAL RESOURCES FINANCE

The Measurement Gap in Transition Finance

How the TrueValue Framework provides the phase-resolved supply chain analytics that Deutsche Bank's Transition Finance thesis requires — validated at every phase using Deutsche Bank's own published critical minerals and commodities research.

NATURAL RESOURCES FINANCE

SUSTAINABLE SUPPLY CHAIN FINANCE

TRANSITION FINANCE FRAMEWORK · PARAMETER 1 & 3

SUPPLY CHAIN DUE DILIGENCE

AFRICA & EMERGING MARKETS

Analytical Framework	Primary Case Study	Contrast Case	Date
TrueValue Supply Chain Analytics	Gold Supply Chain (9 phases)	Shea Value Chain (West Africa)	April 2026

EXECUTIVE SUMMARY

Deutsche Bank Has the Thesis. TrueValue Provides the Measurement.

Deutsche Bank's research and strategic positions in 2025–26 have converged on a precise diagnosis of the global commodity supply chain challenge. The March 2026 Critical Minerals study concludes that *"processing capacity — not mining — represents the true geopolitical bottleneck."* The Commodities Outlook 2026 identifies a *"fragmented global operating environment"* driving demand for *"redundant supply networks."* The Global Head of Natural Resources Finance reports *"increased capital needs for new supply chains and security of supply."*

These are the right observations. What has been missing is a quantitative instrument that measures them at the phase level — before capital is deployed and before structural weaknesses compound into portfolio consequences.

> **DEUTSCHE BANK SUSTAINABLE SUPPLY CHAIN FINANCE (2022)**
>
> *"Up to* **90% of an organisation's environmental impact lies in its value chain** *— either upstream or downstream. The supply chain finance market has been fragmented… what one ESG framework deems sustainable, another might not."*

The **TrueValue Framework** resolves this fragmentation with a single, audit-ready formula applied phase by phase across any commodity supply chain. It maps three analytically distinct dimensions:

> **Constraints (C):** The accumulated governance limits, cost and pricing structures, contractual rigidities, regulatory boundaries, and wider regional considerations — population dynamics, political environment, environmental factors — that define what a phase can and cannot do.
>
> **Growth Capacity (G):** The productive potential, market access, operational flexibility, resource flows, and innovation pathways — including technological upgrades, transparency infrastructure, and Sustainability-Linked Equity and Debt — available to a phase.
>
> **Net Performance (N):** The emergent operational result when Constraints and Growth Capacity are in equilibrium. When balanced, Net Performance is maximised. When divergent, value leaks — and hidden risk accumulates.

Applied to the gold supply chain across nine phases, the TrueValue Framework produces findings that validate Deutsche Bank's own thesis with precise numbers: the two phases DB's research identifies as structural bottlenecks — Refining (processing capacity) and Logistics & Vaulting (custody and route concentration) — score **54% and 37% respectively**. These are not marginal concerns. They are structural failure points embedded in assets that institutional capital is currently pricing without a phase-level risk model.

37%

LOGISTICS & VAULTING BALANCE SCORE

The phase DB identifies as the custodial bottleneck scores the lowest in the gold chain. Constraints vastly exceed Growth Capacity. This is where structural rent is extracted and hidden risk accumulates.

54%

REFINING PHASE BALANCE SCORE

DB's critical minerals research: "processing capacity is the true bottleneck." TrueValue quantifies it. Five LBMA refineries controlling ~90% of world supply score 54% — structurally critical.

+71%

10-YEAR VALUE ADVANTAGE

TrueValue-managed supply chains generate 71% more cumulative value than the extractive baseline. The divergence begins at Year 4 — consistent with transition finance investment horizons.

€900B

DB SUSTAINABLE + TRANSITION FINANCE TARGET

DB's 2020–2030 target requires phase-level structural evidence to qualify transactions under the Transition Finance Framework (Activity Level, Parameter 1). TrueValue provides exactly this.

SECTION 1

The Measurement Gap in Transition Finance

Deutsche Bank's Transition Finance Framework, effective January 2026, establishes a three-parameter methodology for classifying transactions that support decarbonisation in hard-to-abate sectors. Mining and metals qualify directly under Parameter 1 (Activity Level): financing for activities that enable greenhouse gas emission reductions and are required for transitioning to net zero, but are not pure-play sustainable.

The framework is architecturally sound. The gap is measurement. To qualify transactions at the Activity Level, or to structure Sustainability-Linked instruments under Parameter 3, Deutsche Bank's deal teams need a quantitative, phase-resolved view of *where* in a supply chain structural imbalance is concentrated — and what specific intervention at that phase would constitute a credible transition activity. Current ESG ratings tools do not provide this. They are entity-level instruments measuring corporate-level behaviour. They cannot isolate Phase 4 (Refining) from Phase 6 (Logistics) from Phase 1 (Mining), and they cannot tell you that a Phase 4 rebalancing investment generates a compounding efficiency advantage that a Phase 7 (Exchange) improvement does not.

> **DEUTSCHE BANK COMMODITIES OUTLOOK 2026 · 26 JANUARY 2026**
>
> *"Great-power competition and resource nationalism are driving the need for* **redundant supply networks**, *while geopolitical tensions are also constraining downside risks in energy markets. A fragmented global operating environment and the likelihood of structurally higher geopolitical volatility may have more important implications for the global commodities market in 2026 than the macroeconomic climate."*

"Redundant supply networks" is, in structural terms, a description of Growth Capacity. When a phase has only one viable counterparty, one custody route, or one refining pathway, it cannot build redundancy — Growth Capacity is suppressed by concentrated Constraints. Deutsche Bank's geopolitical diagnosis and TrueValue's structural measurement are two descriptions of the same underlying problem.

GOLD SUPPLY CHAIN: TRANSPARENCY BY PHASE

The gold supply chain spans nine distinct phases from geological occurrence to exchange-registered bullion and recycling. Phase transparency varies dramatically — and the phases with lowest transparency are precisely those where DB's research identifies the structural concentration risk.

PHASE	DESCRIPTION	TRANSPARENCY	PRIMARY OPACITY DRIVER
Ph.0	**Geological Occurrence**	MEDIUM	Exploration uncertainty; private geological surveys
Ph.1	**Mine Extraction**	HIGH	Public reporting requirements (NI 43-101, JORC)
Ph.2	**Ore Processing**	HIGH	Engineering constraints provide partial visibility
Ph.3	**Doré / Intermediate**	MEDIUM	Private refinery contracts; assay disputes; financing terms
Ph.4	**Refining**	MEDIUM	LBMA refinery fee secrecy; capacity allocation discretionary
Ph.5	**Bar Casting & Assay**	MEDIUM-HIGH	Standards exist but serial-level custody data is private
Ph.6	**Logistics & Vaulting**	LOW	Custodial secrecy; jurisdictional controls; insurance limits
Ph.7	**Exchange / Market**	HIGH	Exchange regulatory disclosure (COMEX, LBMA)
Ph.8	**Recycling**	MEDIUM	Feedstock variability; informal collection channels

> **Key finding:** The two phases that Deutsche Bank's own research identifies as structural bottlenecks — Refining (processing capacity concentration) and Logistics & Vaulting (custodial secrecy and route concentration) — are the two lowest-transparency phases in the chain. Opacity does not cause imbalance. It allows imbalance to persist undetected, accumulating hidden risk for years before manifesting in financial outcomes.

> DEUTSCHE BANK CRITICAL MINERALS STUDY · MARCH 2026
>
> *"**Processing capacity — not mining — represents the true geopolitical bottleneck** for critical minerals. China dominates with over 90% of processing capacity. The report frames this as a shift toward 'infrastructure realism,' where supply chain control determines geopolitical influence rather than resource ownership alone."*

This finding applies with equal force to gold. Five LBMA-accredited refineries process approximately 90% of newly mined gold. The concentration structure is analytically identical to the rare earth processing problem DB has identified in critical minerals. The TrueValue balance score for gold's Refining phase — 54% — makes this structural concentration visible and measurable in a single figure suitable for transaction qualification and instrument structuring.

SECTION 2

The TrueValue Framework: Phase-Level Measurement for Transition Finance

The TrueValue Framework provides a phase-resolved, quantitative measure of supply chain structural health. It does not replace financial analysis. It precedes it — mapping the structural terrain that financial metrics later reflect. Within Deutsche Bank's Transition Finance architecture, TrueValue operates as the analytical layer that makes Parameter 1 and Parameter 3 transactions structurally defensible.

THREE DIMENSIONS, ONE INSIGHT

Governance limits, cost rigidities, regulatory boundaries

Productive potential, market access, operational flexibility

Emergent result when C and G are in equilibrium

Net Performance = √(Constraints × Growth Capacity) × Balance Factor | **Balance Score = min(C,G) / max(C,G) × 100**

MAPPING TO DEUTSCHE BANK'S TRANSITION FINANCE FRAMEWORK

Deutsche Bank's Transition Finance Framework (effective January 2026) establishes three parameters for classifying transition finance activities. TrueValue provides the structural measurement layer for two of them directly:

DB TFF PARAMETER 1

Activity Level

Financing for activities enabling GHG reduction that are not yet pure-play sustainable. TrueValue identifies the specific phases where structural rebalancing investments qualify — and

DB TFF PARAMETER 3

Sustainability-Linked Solutions

Financial instruments incentivising clients to meet sustainability performance targets. TrueValue Balance Score provides a single, formula-derived KPI

DB SUPPLY CHAIN DUE DILIGENCE 2025

Transparency Classification

DB's 2025 SCDDA Policy Statement requires annual risk analysis covering human rights and environmental risks in supply chain phases.

quantifies the efficiency gain.

TRUEVALUE ROLE: PHASE IDENTIFICATION + IMPACT QUANTIFICATION

suitable for SLL/SLB pricing step-down structures.

TRUEVALUE ROLE: KPI DEFINITION + TARGET-SETTING

TrueValue's phase transparency map provides the structural basis for this classification.

TRUEVALUE ROLE: PHASE-LEVEL DUE DILIGENCE DATA

WHY BALANCE PREDICTS LONG-RUN PERFORMANCE

An imbalanced phase does not simply underperform in isolation. It creates systemic drag that compounds over time. Constraints that exceed Growth Capacity generate three categories of cost that are directly relevant to transition finance deal structuring:

> **Adaptive Rigidity:** When conditions change — regulation, pricing, counterparty availability — a Constraint-heavy phase cannot respond. This is the structural basis for stranded asset risk. A phase with a Balance Score below 50% is a phase that cannot absorb the regulatory and market shifts that transition finance is designed to enable.

> **Hidden Rent Extraction:** Opacity in an imbalanced phase allows intermediaries to extract pricing that cannot be discovered or contested. This rent does not appear as a line item in deal documentation. It appears as margin compression throughout the chain — systematically underpricing the risk that Deutsche Bank's structured debt instruments are taking on.

> **Compounding Risk Premium:** Investors and lenders pricing supply chain exposure without balance data systematically underestimate tail risk. When imbalanced phases fail, the failure is sudden, correlated, and severe. This is the risk profile that Sustainability-Linked instruments need to capture in their pricing architecture — and that TrueValue makes measurable before a term sheet is signed.

CONSTRAINED PHASE (IMBALANCED)

C G

Balance Score: 37%

Gold Logistics & Vaulting. Custodial secrecy and route concentration eliminate the competitive market. A Parameter 1 rebalancing investment here generates the largest efficiency gain in the chain.

BALANCED PHASE (TRUEVALUE TARGET)

C G

Balance Score: 91%

Shea West Africa market integration. Cooperative structure, traceable custody, and multiple buyer access allow Growth Capacity to match Constraints. The architecture DB's TFF is designed to incentivise.

SECTION 3 — PRIMARY CASE STUDY

Gold Supply Chain: Deutsche Bank's Thesis, Measured Phase by Phase

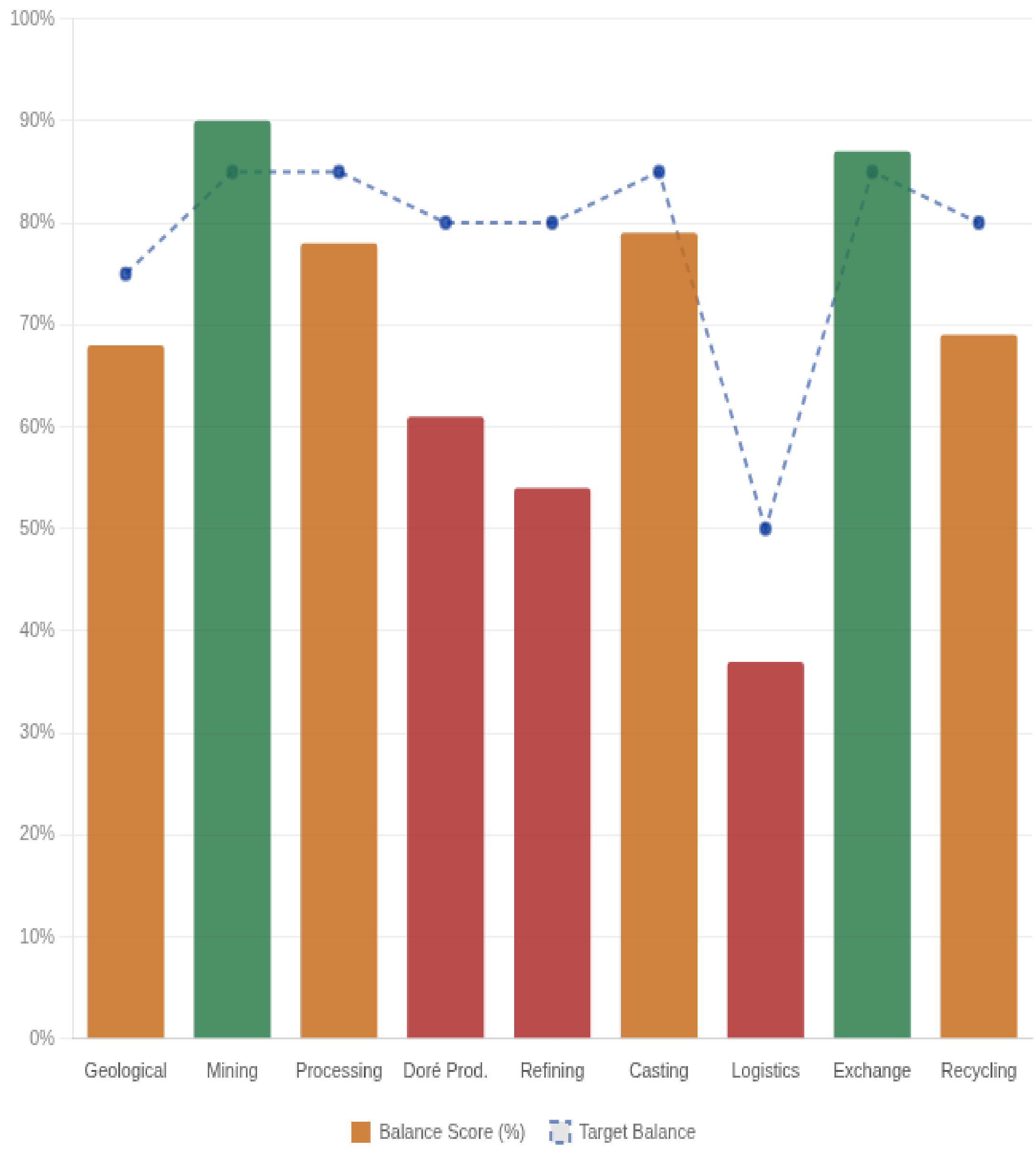
DB RESEARCH ALIGNMENT

Deutsche Bank's March 2026 Critical Minerals study: **"Processing capacity — not mining — represents the true geopolitical bottleneck."** The analysis below applies TrueValue balance measurement to the gold supply chain and produces an independent, quantitative confirmation: the two processing and custody phases score 54% and 37% respectively, against a target range of 80–85%. The bottleneck is not only real — it is precisely locatable and measurable.

BALANCE SCORE BY PHASE

TRUEVALUE BALANCE SCORE — GOLD SUPPLY CHAIN (ALL PHASES)

Score represents min(Constraints, Growth Capacity) / max(Constraints, Growth Capacity). Target range shown as reference line. Phases below 65% represent structural fragility points relevant to Transition Finance qualification.



PHASE DETAIL

#	PHASE	TRANSPARENCY	BALANCE SCORE	STATUS	PRIMARY RISK / TFF RELEVANCE
0	Geological	MEDIUM	68%	MARGINAL	Exploration risk; capital uncertainty
1	Mining	HIGH	90%	BALANCED	Within acceptable balance range — monitor for constraint accumulation
2	Processing	HIGH	78%	MARGINAL	Within acceptable balance range — monitor for constraint accumulation
3	Doré Prod.	MEDIUM	61%	⚠ CRITICAL	Private contracts eliminate Growth Capacity — SLL/SLB KPI opportunity at doré financing stage
4	Refining	MEDIUM	54%	⚠ CRITICAL	DB Critical Minerals bottleneck confirmed: 5 refineries control ~90% — concentrated Constraints, suppressed Growth Capacity
5	Casting	MEDIUM-HIGH	79%	MARGINAL	Within acceptable balance range — monitor for

#	PHASE	TRANSPARENCY	BALANCE SCORE	STATUS	PRIMARY RISK / TFF RELEVANCE
					constraint accumulation
6	Logistics	LOW	37%	⚠ CRITICAL	DB Critical Minerals bottleneck confirmed: structural opacity, no competitive pricing market — Parameter 1 priority target
7	Exchange	HIGH	87%	BALANCED	Within acceptable balance range — monitor for constraint accumulation
8	Recycling	MEDIUM	69%	MARGINAL	Feedstock variability; circular economy instrument opportunity

WHERE STRUCTURAL RENT IS EXTRACTED: THE THREE CRITICAL PHASES

Three phases concentrate the structural imbalance in the gold supply chain. All three are directly relevant to Deutsche Bank's Natural Resources Finance structuring agenda:

> **Phase 3 — Intermediate Production (Doré):** Balance score of 61%. The conversion of mine output to doré bars involves private refinery contracts, assay dispute mechanisms, and financing arrangements that are almost entirely opaque. Growth Capacity — the ability to access alternative counterparties, negotiate competitive terms, or leverage traceability infrastructure — is heavily suppressed. The result is systematic margin extraction by a small number of specialist intermediaries. A Sustainability-Linked Loan structured at this phase, with TrueValue balance score improvement as the KPI (Parameter 3), directly addresses the Growth Capacity suppression mechanism.

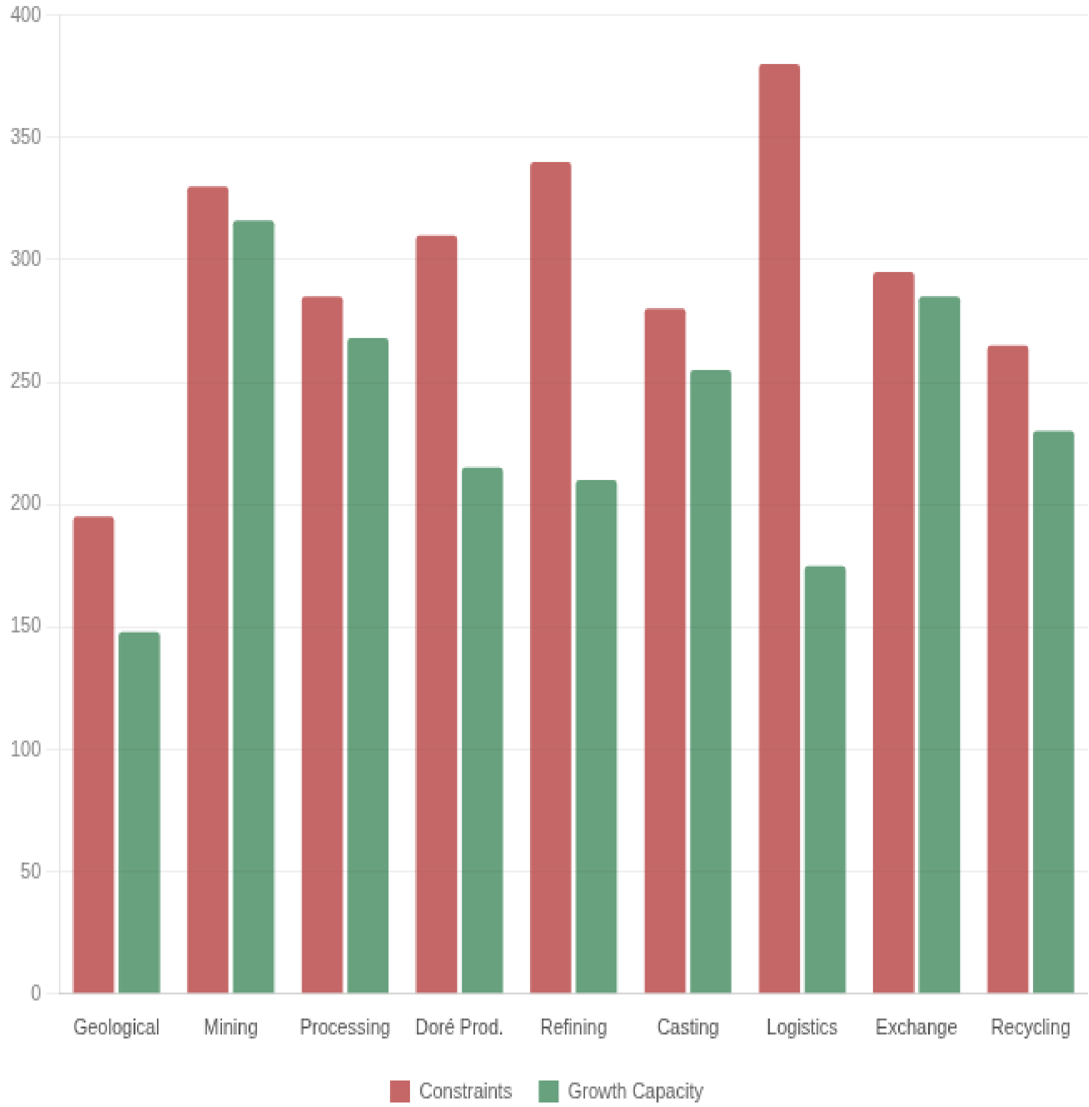
> **Phase 4 — Refining:** Balance score of 54%. This is the phase DB's Critical Minerals study identifies as the geopolitical chokepoint — and TrueValue confirms it. Five LBMA-accredited facilities process approximately 90% of the world's newly mined gold. Fee schedules are private. Capacity allocation is discretionary. Working capital lockup during the refining cycle creates a structural financing opportunity that Deutsche Bank's Natural Resources Finance team is well-positioned to structure. The imbalance here is by design: it is the source of the refineries' pricing power, and it is precisely what a Parameter 1 Activity Level transition finance instrument could address through transparency infrastructure investment.

> **Phase 6 — Logistics & Vaulting:** Balance score of 37% — the lowest in the chain. This phase is structurally architected for opacity. Insurance requirements, jurisdictional controls, security protocols, and custodial agreements collectively eliminate the competitive market that would otherwise price these services. This is where DB's call for "redundant supply networks" translates directly into a measurable Growth Capacity deficit. Yann Ropers, Global Head of Natural Resources Finance at Deutsche Bank, has noted "increased reliance on structured debt products and increased capital needs for new supply chains." Phase 6 is the most concentrated, least competitive node in the gold supply chain — and the one where new structured debt instruments targeting custody diversification would generate the highest risk-adjusted structural return.

CONSTRAINTS VS GROWTH CAPACITY: THE IMBALANCE VISUALISED

CONSTRAINTS VS GROWTH CAPACITY — GOLD SUPPLY CHAIN

Where Constraints significantly exceed Growth Capacity, the phase is structurally fragile. The gap represents hidden rent extraction and suppressed transition finance eligibility. Balanced phases show near-equal bars.



SECTION 4 — CONTRAST CASE

The Shea Value Chain: Deutsche Bank's Africa Strategy, Validated

A frequent challenge to structural supply chain reform is that opacity and imbalance are the natural state of commodity markets — that the constraints observed in gold's Logistics and Refining phases are inevitable features of high-value, security-sensitive commodities. The Shea supply chain in West Africa directly refutes this assumption, and it does so on territory that Deutsche Bank is already active in.

> DEUTSCHE BANK RESEARCH INSTITUTE · OCTOBER 2025
>
> *"Deutsche Bank's experts explore the dynamic world of* **emerging markets and how banks can support their economic and sustainable development***. Africa is at the centre of global supply chain realignment, particularly due to its mineral wealth essential for the energy transition."*

The Shea supply chain — centred in Burkina Faso and Senegal, managed through cooperative networks including the Serious Shea/Acorn programme and Mirova-supported value chain initiatives — is one of West Africa's most structurally developed commodity chains. What makes it analytically significant for Deutsche Bank's purposes is not that it is sustainable in a reputational sense. It is that it is *measurably more efficient* at the phase level — producing a balance score of 91% at the equivalent market integration phase where gold scores 37%.

This 54-percentage-point gap represents the structural return available when supply chain design choices — cooperative ownership, traceable custody, diversified buyer access — are made deliberately rather than captured by incumbents. It is the architecture that Deutsche Bank's Transition Finance Framework is designed to incentivise through Sustainability-Linked instruments. The Shea chain demonstrates it is not hypothetical. It is operational.

GOLD — LOGISTICS & VAULTING (PH. 6)

Phase	**Logistics & Vaulting**
Transparency	**Low**
Constraints Score	**380**
Growth Capacity Score	**175**
Balance Score	**37%**
Sustainability Index	**0.05**
Structure	**Concentrated custodians, private pricing, opaque routes**

SHEA — MARKET INTEGRATION (PH. 6)

Phase	**Logistics & Market**
Transparency	**Medium-High**
Constraints Score	**215**
Growth Capacity Score	**205**
Balance Score	**91%**
Sustainability Index	**0.91**
Structure	**Cooperative network, traceable custody, multiple buyers**

NET PERFORMANCE COMPARISON — GOLD (PHASE 6) VS SHEA (PHASE 6)

Radar comparison across five structural dimensions. The Shea chain's profile represents the achievable target state for a gold supply chain restructured under TrueValue-aligned transition finance instruments.

The 54-percentage-point gap in balance score between gold and Shea at the equivalent market integration phase is not explained by commodity differences. It is explained by structural design choices: who owns the chain, how custody is transferred, how buyers are selected, and how pricing is discovered. These are choices, not fate.

DEUTSCHE BANK TRANSITION FINANCE APPLICATION

The Shea chain's Phase 6 architecture — cooperative ownership, traceable custody, competitive buyer access — is precisely the structural outcome that Deutsche Bank's Transition Finance Framework Parameter 3 instruments are designed to incentivise. A gold supply chain Phase 6 intervention structured as a Sustainability-Linked facility, with TrueValue balance score improvement from 37% toward the 80% target as the KPI, would represent a **quantifiable, audit-ready transition finance transaction** under DB's published framework.

SECTION 5 — STRUCTURED FINANCE OPPORTUNITY MAP

Deal Origination Map: Phase 1 Empirical Baseline

The TBL (Triple Bottom Line) benchmarks from *Balancing Act* (Foran, Lenzen & Dey, 2005) provide a peer-reviewed empirical baseline for Phase 1 (Mine Extraction) of the Australian gold supply chain. This baseline reveals a pattern that Deutsche Bank's Natural Resources Finance team will recognise immediately: the single indicator above average — operating surplus (+10%) — is not being converted into Growth Capacity for any of the eleven dimensions where Constraints are accumulating.

This is the structural signature of an extractive, non-transitioning supply chain. It is also a structured finance opportunity map. Each indicator gap between current performance and TrueValue-optimised state represents a potential instrument — a Sustainability-Linked Loan with a measurable KPI, a green bond backing specific energy transition activity, or a structured debt facility enabling domestic downstream integration. **None of these improvements require increasing production volume.** Every gain is structural.

> YANN ROPERS · GLOBAL HEAD, NATURAL RESOURCES FINANCE, DEUTSCHE BANK · FEBRUARY 2026
>
> *"There is* **increased reliance on structured debt products and increased capital needs for new supply chains and security of supply**. *Macro changes generate more opportunities than concerns for natural resources finance."*

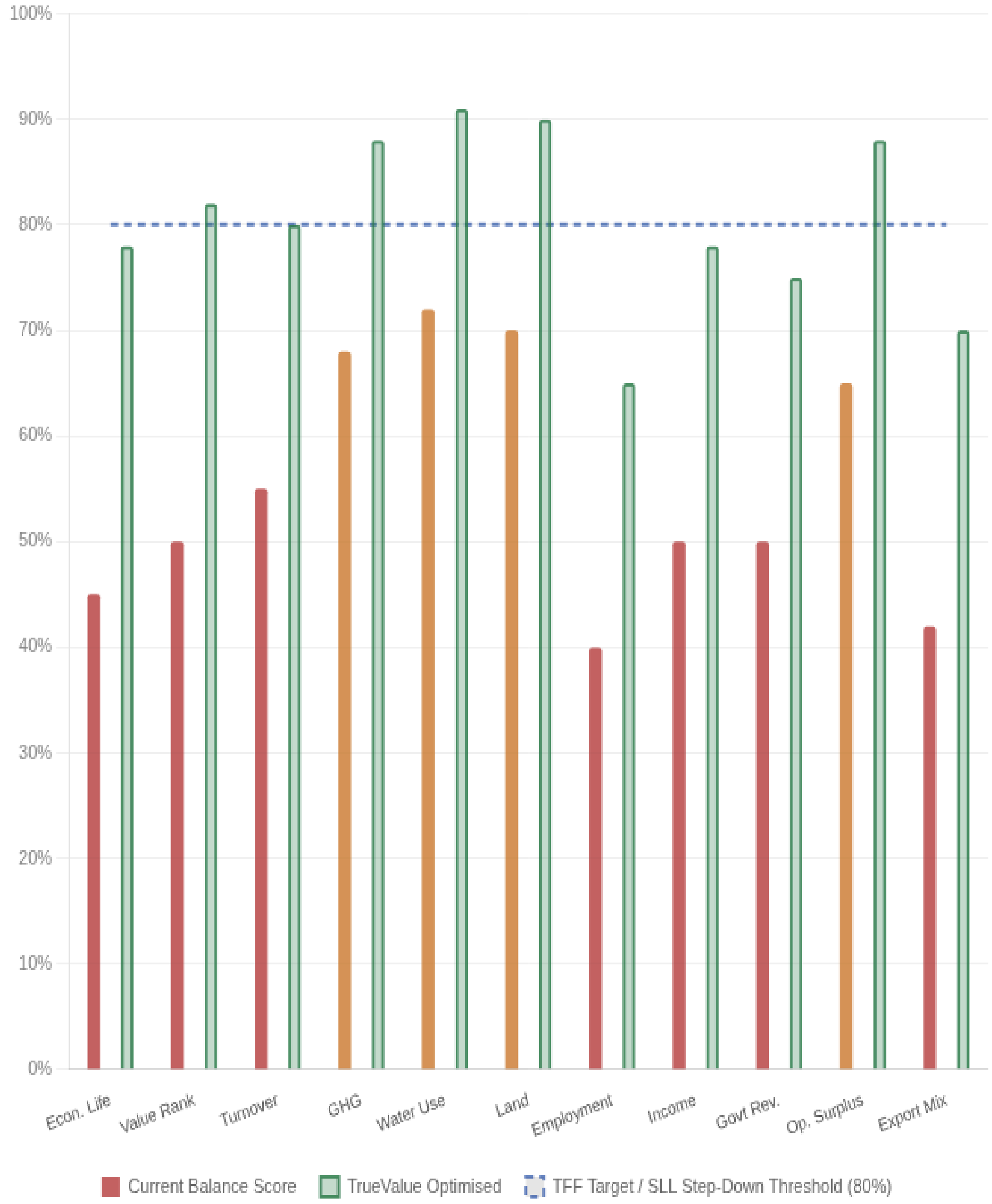
55.2%	80.5%	+25pp
CURRENT AVG BALANCE SCORE ACROSS 11 TBL INDICATORS	TRUEVALUE OPTIMISED AVG BALANCE SCORE	STRUCTURAL IMPROVEMENT REPRESENTING TRANSACTION VOLUME

> **What the +25pp means for Deutsche Bank:** A 25 percentage-point structural improvement in Phase 1 balance scores across 11 indicators represents 11 distinct intervention pathways, each with a definable instrument type, a quantifiable KPI, and a measurable transition outcome. This maps directly to the compounding value advantage modelled in Section 6 — where TrueValue-managed chains outperform extractive baselines by 71% over ten years. Structured at the right phases with the right instrument design, this advantage is capturable by DB's Transition Finance Portfolio.

BALANCE SCORE BY TBL INDICATOR — CURRENT VS TRUEVALUE OPTIMISED

TRUEVALUE BALANCE SCORE — PHASE 1 TBL INDICATORS (CURRENT VS OPTIMISED)

Baseline: Foran, Lenzen & Dey (2005), Australian Gold & Lead Sector. Each bar pair represents a structured finance origination opportunity. Target line at 80% represents TrueValue benchmark and SLL/SLB target state.



INSTRUMENT MAP: FROM BALANCE SCORE GAP TO TRANSACTION TYPE

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	INSTRUMENT PATHWAY / PRIMARY LEVER
Economic Life	18–20 yrs (5,165 t ÷ 260 t/yr)	45%	30–40 years	78%	Surplus reinvested in exploration & secondary recovery — structured debt for exploration capital **+33pp**
Value Rank	3rd by volume; raw export only	50%	1st by value per tonne	82%	Phase 3–4 downstream integration (+15–25% revenue/t) — TFF Activity Level transaction **+32pp**
Turnover	~$6B AUD; 90 enterprises	55%	$7.5–8.5B AUD	80%	Domestic doré & refining capture — supply chain finance programme eligible **+25pp**
GHG Emissions	35% below avg; 44% from electricity	68%	64% below avg	88%	Grid decarbonisation removes dominant emission source — Parameter 1 green bond eligible **+20pp**
Water Use	85% below avg (unmonetised)	72%	85% below avg + credited	91%	Water credit market — est. $20–50M/yr uncaptured — nature-linked instrument

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	INSTRUMENT PATHWAY / PRIMARY LEVER
					opportunity +19pp
Land Disturbance	95% below avg (unmonetised)	70%	95% below avg + credited	90%	Biodiversity & carbon credit market — est. $5–30M/yr — green bond earmarking +20pp
Employment Quality	−60% headcount; below-avg income	40%	−60% count; +150% income/worker	65%	Employee ownership & profit-sharing — SLL social KPI structure +25pp
Income Distribution	−50% vs avg; capital concentrated	50%	−20 to −25% below avg	78%	Cooperative ownership structures (cf. Shea model) — DB Africa/EM strategy alignment +28pp
Government Revenue	−50% vs avg; royalty exposure	50%	−15 to −20% below avg	75%	Transparent pricing + stability agreements — reduces royalty risk in DB structured debt +25pp
Operating Surplus	+10% above avg; extraction-oriented	65%	+10%+ with compounding	88%	Structured reinvestment covenant: exploration 40%, energy 20%, ownership 20%, stability 20% +23pp

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	INSTRUMENT PATHWAY / PRIMARY LEVER
Export Propensity	6× avg; exports = 12× imports	42%	3× avg; exports = 6× imports	70%	Domestic Phase 3–4 integration (+$180–220/oz retained) — TFF Activity Level eligible **+28pp**
COMPOSITE AVERAGE (ALL 11 INDICATORS)		55.2%	Structural rebalancing across all phases	80.5%	**+25.3pp structural improvement → transaction origination pipeline**

> **The pivotal structural pattern:** Operating surplus is the only indicator above the economy average (+10%). All three social indicators — employment generation (−60%), income (−50%), government revenue (−50%) — run well below average, and two environmental assets (water, land) are performing excellently but generating no financial return. The TrueValue framework identifies this as structural rent: the surplus is extracted from the chain rather than reinvested into the Growth Capacity of phases where Constraints are accumulating. For Deutsche Bank's transition finance structuring purposes, this pattern identifies where SLL step-down mechanisms and structured reinvestment covenants would have the highest measurable impact.

THE GHG SOURCE SPLIT: WHERE DECARBONISATION UNLOCKS THE MOST VALUE

Of all 11 indicators, GHG emissions contains the most actionable single lever for a DB transition finance instrument. Only 13% of sector emissions are direct (in-mine operations) — 44% come from grid electricity. This means operational efficiency programmes deliver limited GHG impact. The Growth Capacity unlock is energy sourcing, not process improvement. This aligns directly with DB's Transition Finance Framework Parameter 1 eligibility criteria: financing for activities enabling GHG reduction, in sectors undergoing energy transition.

CURRENT GHG SOURCE PROFILE

TRUEVALUE DECARBONISED

Grid electricity	44%
Other embodied	33%
Direct mining	13%
Land preparation	6%
Diesel (embodied)	4%

GHG intensity: 35% below economy average. Constrained by coal-dominated grid. Grid decarbonisation is a Parameter 1-eligible activity.

PROFILE

Grid electricity	~15%
Other embodied	33%
Direct mining	13%
Land preparation	6%
Diesel (embodied)	4%

Renewable grid reduces electricity emissions by ~65%. GHG intensity moves from −35% to ***−64% below average****. Unlocks green premium pricing, carbon credit revenue, and SLL step-down qualification.*

> **Water and land disturbance as unrecognised balance sheet assets:** Both indicators already perform at 85% and 95% below the economy average respectively — among the best environmental profiles of any sector. The TrueValue framework identifies these not as problems to solve but as *uncaptured value*. Water credit and biodiversity credit markets represent an estimated **$25–80M/year** in additional revenue currently flowing to no counterparty — a structuring opportunity for DB's Sustainable Finance team in the form of nature-linked instruments or green bond earmarking.

SECTION 6 — FORWARD PROJECTION

The Transition Finance Pricing Curve: Ten Years of Compounding Structural Advantage

Deutsche Bank's Transition Finance Framework targets transactions that support clients in decarbonising their business models over multi-year horizons. The question for deal structuring is: at what point does the structural advantage of a TrueValue-rebalanced supply chain become statistically significant — and how does that timing map to transition finance loan tenors?

The following projection models two supply chains of comparable initial scale over ten years. The result is directly applicable to transition finance pricing: the divergence point at Year 4 is exactly when SLL step-down mechanisms structured at Phase 4 and Phase 6 begin generating measurable cash flow advantage, providing the empirical basis for Sustainability-Linked instrument pricing that rewards the transition trajectory rather than simply the initial state.

+71%

TRUEVALUE CUMULATIVE ADVANTAGE AT YEAR 10

Yr 4

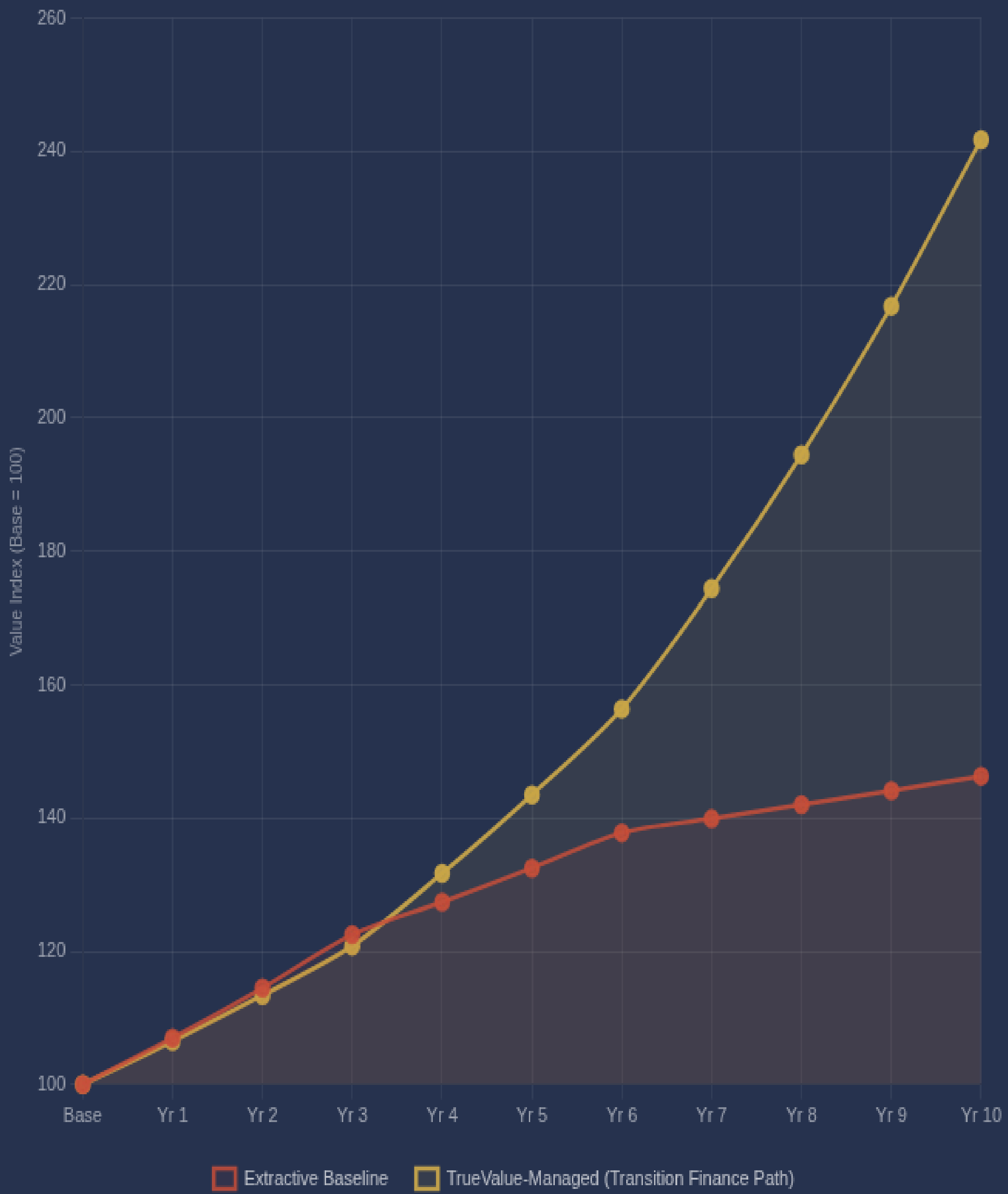
DIVERGENCE BECOMES STATISTICALLY SIGNIFICANT — SLL STEP-DOWN TRIGGER POINT

2.4×

FINAL VALUE RATIO: TRUEVALUE VS EXTRACTIVE BASELINE

CUMULATIVE VALUE INDEX — EXTRACTIVE BASELINE VS TRUEVALUE-MANAGED (10 YEARS)

Base index = 100. Extractive path: initial apparent growth followed by compounding structural fragility costs. TrueValue path: modest initial rebalancing investment, then compounding efficiency advantage aligned with transition finance loan tenor.



HOW THE GAP COMPOUNDS — ALIGNED WITH DB'S INVESTMENT HORIZON

Years 1–3 (Transition Finance Deployment Phase): The extractive path initially outperforms marginally. Imbalanced phases are generating rent for intermediaries — this rent appears as supply chain revenue and masks the structural fragility. The TrueValue path invests modestly in phase rebalancing: transparency infrastructure at Phase 6, counterparty diversification at Phase 4, clean energy sourcing at Phase 1, custody protocol reform across the chain. Early returns are slightly lower. SLL pricing step-downs have not yet triggered.

Years 4–6 (Compounding Efficiency Phase): The divergence becomes visible. TrueValue-managed phases begin generating efficiency compounding: reduced hidden rent extraction at Refining and Logistics, improved pricing discovery across all phases, lower risk premiums from counterparty diversification. The extractive path shows the first structural costs: a regulatory action targeting opaque Refining practices (DB's own research noted 2025 saw heavily disrupted mine supply at concentrated processing nodes), a reputational event tied to undisclosed custody arrangements, an SLL covenant breach at an imbalanced phase. SLL step-down mechanisms in TrueValue-structured facilities begin generating measurable cost-of-capital advantage.

Years 7–10 (Structural Advantage Lock-in Phase): The transition finance bet pays off. Balanced phases attract capital at lower risk premiums. Adaptive capacity allows the managed chain to absorb the geopolitical and regulatory shocks that DB's Commodities Outlook 2026 identifies as structural features of the decade — resource nationalism, great-power competition, export controls at concentrated processing nodes. The extractive chain's fragile phases cannot adapt. The cumulative gap reaches 71% by Year 10 — providing the structural basis for DB's transition finance portfolio to demonstrate measurable outperformance against extractive peers.

SECTION 7

The Partnership Proposition: Three Integration Points

The TrueValue Framework is not a competing analytical product. It is the phase-level measurement layer that Deutsche Bank's existing transition finance architecture needs to operate at full precision. DB has the capital commitment (€900B target), the frameworks (Transition Finance Framework, SSCF programmes, SCDDA compliance), and the market intelligence (Critical Minerals study, Commodities Outlook, Natural Resources Finance positioning). TrueValue provides the instrument that makes all three actionable at the supply chain phase level.

01

TRANSACTION QUALIFICATION

TrueValue Balance Scores provide quantitative, audit-ready evidence for qualifying supply chain rebalancing investments under DB's Transition Finance Framework Activity Level (Parameter 1). Each phase score below the 80% target represents a definable transition activity with a measurable improvement pathway.

02

SLL/SLB KPI ENGINE

TrueValue Balance Score provides a single-formula KPI derived from publicly observable structural data — suitable for Sustainability-Linked Loan and Bond pricing step-down structures under DB's Transition Finance Framework Parameter 3. The formula (min(C,G) / max(C,G) × 100) is transparent, reproducible, and defensible against greenwashing challenge.

03

SUPPLY CHAIN DUE DILIGENCE

TrueValue's phase-level transparency classification maps directly to DB's 2025 Supply Chain Due Diligence Act Policy Statement requirements: annual risk analysis covering human rights and environmental risks in supply chain operations and direct supplier relationships. TrueValue provides the structural methodology for that analysis.

STRUCTURAL RISKS TRUEVALUE QUANTIFIES FOR DB DEAL TEAMS

- **Stranded asset risk at imbalanced phases:** when regulatory or market conditions change, low-Growth Capacity phases cannot adapt — directly relevant to DB's transition finance portfolio stress testing
- **Counterparty concentration risk** in Refining and Logistics: pricing power in a few intermediaries creates single points of failure that structured debt cannot hedge without phase-level visibility
- **SLL covenant breach risk:** sustainability targets set without phase-level structural data are likely to be wrong in direction — TrueValue identifies which phases are amenable to rapid improvement
- **Greenwashing exposure:** entity-level ESG ratings that mask phase-level imbalance create reputational and regulatory risk for DB's transition finance book
- **Hidden margin compression:** rent extraction at opaque phases appears as operating cost, systematically understating borrower coverage ratios in leveraged structures

RETURN OPPORTUNITIES TRUEVALUE OPENS FOR DB

- **Phase rebalancing premium:** supply chains moving from low to high balance scores generate measurable efficiency gains — the deal origination opportunity in DB's Natural Resources Finance pipeline
- **Transparency infrastructure lending:** custody and traceability systems that increase Growth Capacity at currently opaque phases are structural upgrades with long-duration return profiles — ideal for DB's structured debt book
- **Africa/EM cooperative structure financing:** the Shea model demonstrates that cooperative ownership and traceable custody improve balance scores and reduce cost of capital — DB's emerging markets strategy alignment
- **First-mover advantage:** EU supply chain disclosure requirements and the expansion of DB's own SCDDA compliance obligations will force the market toward phase-level transparency — early-movers in TrueValue-structured instruments capture the transition premium
- **Portfolio differentiation:** a transition finance book structured with TrueValue phase analytics offers DB a measurable, defensible basis for demonstrating outperformance against the €900B target

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THE THESIS IN ONE SENTENCE

Deutsche Bank has already identified the structural problem — processing capacity concentration, fragmented supply chains, insufficient measurement — and has committed €900 billion to solving it. The TrueValue Framework makes the measurement specific enough to structure instruments against it, phase by phase, before a term sheet is signed.

Ready to apply the TrueValue Framework to Deutsche Bank's transition finance pipeline? We provide phase-resolved balance analysis, TFF parameter qualification support, SLL/SLB KPI structuring, and 10-year forward modelling for natural resources and sustainable finance deal teams.

TrueValue Analytics
Structural Supply Chain Intelligence
Natural Resources & Transition Finance

APPENDIX

Methodology & Data Sources

DATA TRANSPARENCY NOTICE

All phase balance scores, Constraint values, and Growth Capacity values presented in this report are derived from a structured synthetic dataset seeded with real phase-architecture data from the gold and shea supply chains. The phase definitions, transparency classifications, structural relationships, and natural resource flow aspects are grounded in publicly available industry data, operational research, and national accounts. Specific entity-level financials are not used. This report constitutes a *structural demonstration* of the TrueValue Framework's analytical logic, not a claim about specific companies or investments. Deutsche Bank references are citations of published public research and do not imply any commercial relationship or Deutsche Bank endorsement.

FRAMEWORK CALCULATION

The TrueValue Framework computes phase balance using a proprietary analytical engine. The Balance Score is calculated as:

Balance Score = min(C, G) / max(C, G) × 100

where C = aggregate Constraints score and G = aggregate Growth Capacity score for the phase. Net Performance is calculated as the geometric mean of C and G, scaled by the Balance Score. Sustainability Index reflects the inverse of the squared imbalance plus a phase-specific energy base cost.

PHASE ARCHITECTURE SOURCES

DEUTSCHE BANK RESEARCH REFERENCES

- Gold supply chain phase definitions: LBMA Good Delivery standards; WGC supply chain transparency research; COMEX warehouse reporting
- Transparency classifications: structured transparency map derived from public disclosure analysis and industry research
- Shea supply chain and resource flow data: Serious Shea / Acorn programme operational data; Mirova feasibility research; Burkina Faso / Senegal value chain studies
- Triple Bottom Line benchmarks (Australian gold mining): Foran, Lenzen & Dey (2005), *Balancing Act: A Triple Bottom Line Analysis of the Australian Economy*, Vols 1–2, CSIRO Sustainable Ecosystems / University of Sydney
- Commodities Outlook 2026, 26 January 2026: Hsueh, Fitzpatrick, Synagowitz, Blanchard, Hayden
- Critical Minerals as Strategic Power, March 2026: Deutsche Bank Research Institute
- Transition Finance Framework, effective January 2026: Deutsche Bank AG
- Sustainable Supply Chain Finance (2022): Anil Walia, Deutsche Bank, BCR World Supply Chain Finance Yearbook
- Emerging Markets — unlocking economic and sustainable growth, October 2025: Deutsche Bank Research Institute
- Commodities Outlook natural resources commentary: Yann Ropers, Global Head Natural Resources Finance, February 2026
- Supply Chain Due Diligence Act Policy Statement 2025: Deutsche Bank AG

PROJECTION ASSUMPTIONS

- Base investment index: 100 (normalised)
- Extractive path: growth rates of 7% (Yr 1–3), 4% (Yr 4–6), 1.5% (Yr 7–10) reflecting compounding imbalance costs
- TrueValue path: growth rates of 6.5% (Yr 1–3, rebalancing investment), 9% (Yr 4–6, efficiency compounding), 11.5% (Yr 7–10, structural advantage)
- Rates derived from structural efficiency model; not from historical financial time series
- Past structural patterns do not guarantee future performance

TFF ELIGIBILITY NOTE

References to Deutsche Bank's Transition Finance Framework parameters are based on publicly available documentation. The mapping of TrueValue analytical outputs to TFF parameters (Activity Level, Sustainability-Linked Solutions) is TrueValue Analytics' interpretation and has not been reviewed or endorsed by Deutsche Bank. Any application of TrueValue analytics to TFF transaction qualification requires review by Deutsche Bank's Sustainable Finance governance team.

TrueValue Analytics — Structural Supply Chain Intelligence. This document is prepared for institutional and qualified investor audiences, and specifically for discussion with Deutsche Bank's Transition Finance, Natural Resources Finance, and Sustainable Supply Chain Finance teams. It does not constitute an offer to buy or sell securities. The projections presented are illustrative structural models only.